

**The University of Hong Kong
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE**

COMP7103A DATA MINING

Date: May 7, 2018

Time: 6:30pm – 8:30pm

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL 7 questions.

1. (10%) A scientist has a collection of bird samples for analysis. Each bird is associated with the following attributes: *weight*, *wing-span*, *beak size*, *lat-long* (the latitude and longitude of the location where the sample was obtained), and *color*. Describe how each attribute could be represented. Also describe a quantitative measure that measures the similarity of two birds.
2. (10%) What is “dimension reduction”? Why dimension reduction is sometimes needed in data analysis? Briefly describe two dimension reduction techniques.
3. (10%) Explain the difference between *Information Gain* and *Gain Ratio* in decision tree classifier construction. Give an advantage of gain ratio over information gain.
4. (10%) For each of the following statements, state whether the statement is TRUE or FALSE. For each false statement, point out the incorrectness.
 - (a) A minimum support threshold can be used to help discretize a numerical attribute to alleviate the fragmented rule problem in quantitative association rule mining.
 - (b) Predicting the future value of a stock, based on its past values is a classification task.
 - (c) Given two binary vectors x_1 and x_2 , the similarity between x_1 and x_2 based on simple matching coefficient is always less than that based on the Jaccard coefficient.
 - (d) All maximal frequent itemsets are closed itemsets.
 - (e) The training error of a decision tree is typically smaller than its generalization error.

5. Consider the set of 10 labeled records shown in Table 1. The records are partitioned into two partitions for the purpose of performing 2-fold cross-validation.

ID	a_1	a_2	a_3	Class
<i>Partition 1</i>				
1	H	X	T	+
2	H	Y	T	-
3	L	X	T	-
4	L	X	F	+
5	L	Y	F	+
<i>Partition 2</i>				
6	H	X	T	+
7	H	X	F	-
8	H	Y	F	-
9	L	Y	T	-
10	L	Y	F	+

Table 1: Dataset for classification

- (a) (5%) Calculate the information gain of splitting Partition 1 using Attribute a_1 as the splitting attribute (i.e., using a_1 to derive a test).
- (b) (15%) Consider the following approach of building a two-level decision tree:
- Use a_1 as the first splitting attribute.
 - Use classification error as the impurity measure to choose the best remaining attribute for splitting at each of the two successor nodes. Attribute “ID” needs not be considered.

Perform 2-fold cross-evaluation on this method using the given dataset and partitioning. Show all decision trees built and calculate the F-measure of Class “+”.

6. Consider the market basket dataset shown in Table 2. There are 8 transactions and 4 items. The number shown under each item in a transaction refers to the quantity of the item sold in the transaction.

Transaction ID	Items			
	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>
1	2	1	0	0
2	1	2	1	0
3	2	0	2	0
4	3	2	3	2
5	0	3	2	0
6	0	3	3	3
7	0	1	2	1
8	0	0	1	2

Table 2: Market basket transactions

- (a) (6%) Given a support threshold $\rho_s = 30\%$ and a confidence threshold $\rho_c = 70\%$, find all frequent itemsets using the Apriori Algorithm. Show all candidate itemsets and frequent itemsets found.
- (b) (3%) List all maximal frequent itemsets.
- (c) (3%) List all valid association rules (those that satisfy the confidence requirement) that are derived from the frequent 3-itemsets found.
- (d) (4%) Compute the lifts of the rules found in Part (c). Which of the rules is most useful according to the result?
- (e) (4%) Table 3 shows the profit of selling each of the items in the dataset. Calculate the utility of all frequent 3-itemsets found in Part (a).

Item	Profit
<i>a</i>	1
<i>b</i>	5
<i>c</i>	2
<i>d</i>	3

Table 3: Unit-profit for each item

7. Table 4 shows the proximity matrix of 4 objects.

	O_1	O_2	O_3	O_4
O_1	0	0.3	0.5	0.1
O_2	0.3	0	0.6	0.4
O_3	0.5	0.6	0	0.2
O_4	0.1	0.4	0.2	0

Table 4: Proximity matrix of 4 objects

- (a) (4%) Is the proximity measure in the given matrix a similarity measure or a dissimilarity measure? Explain your answer.
- (b) (4%) Find the medoid of the 4 objects.
- (c) (6%) Perform single-link hierarchical clustering on the objects. Show your results by drawing a dendrogram. Mark the dendrogram with the proximity values of the merge points.
- (d) (6%) Perform complete-link hierarchical clustering on the objects. Show your results by drawing a dendrogram. Mark the dendrogram with the proximity values of the merge points.

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